

Proposal 9 : Prediction of a student's success

Project:

Building a system to predict if a student can have good academic performance or not.

Dataset:

Kaggle open dataset (<https://www.kaggle.com/datasets/algozee/student>)

This dataset contains detailed information of 1000 students with different factors(academic history, study habits, lifestyles etc.) that may influence their overall academic performance.

Proposed Solution and Real world Application :

We propose to conduct an EDA(exploratory data analysis) to find what are the most important factors that effect a student's overall grade. During our analysis, we are going to use different types of regression models(Linear, Logistic, Lasso etc.) to fit the original data and visualize the results through plots. After finding the vital factors, we will train a model that could predict if a student will perform good or bad based on the information that it accepts. We plan to use models such as random forest, KNN and gradient boosting and then give suggestions to a student if the system assesses his/her grade is relatively low.

This system has the potential of being used in real world as the more detailed information we feed to it, the more accurate it can predict if a student will perform good or bad earlier and can generate useful suggetions to enhance students' performances.

Project steps

Step	Estimated completion time(tentative)	People in charge
1. EDA, regression analysis, visualization	One week	Zichen Hu, Xiaobo Li
2. Prediction system	One week	Mingke Wei, You Chen
3. Evaluation, report, presentation	One week	Kiarash Kianfar, Zichen Hu